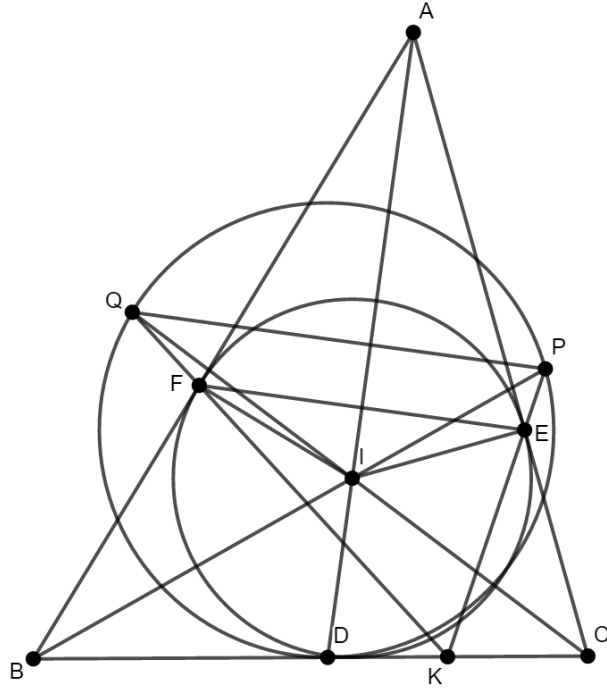


4.3.28



Since P is the intersection of the angle bisector of $\angle ABD$ and the perpendicular bisector of AD , P lies on $\odot ABD$. Similarly, Q lies in $\odot ACD$. Obviously, $EF, PQ \perp AI$, so $EF \parallel PQ$. By simple angle chasing, $\odot DPQ$ is tangent to BC . Then

$$\frac{d(P, BC)}{d(Q, BC)} = \frac{\cos^2 \frac{B}{2}}{\cos^2 \frac{C}{2}} = \frac{d(E, BC)}{d(F, BC)}$$

so PE and QF meets on BC .